

NORMAN NWS, OK, USA

WMO: 723570

Lat: 35.1808N

Lon: 97.4378W

Elev: 1182

StdP: 14.08

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 03948

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.9	19.3	-0.2	5.7	19.7	7.1	8.3	23.7	28.3	44.6	26.0	45.3	9.6	10	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	101.5	74.8	99.1	75.1	95.4	75.5	79.1	92.8	78.1	91.3	77.2	89.9	10.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	139.0	83.3	74.7	135.9	83.0	73.2	129.0	82.3	43.4	92.4	42.4	91.6	41.4	89.4	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.4	23.1	20.3	8.8	103.7	4.4	4.6	5.6	107.0	3.0	109.7	0.5	112.3	-2.7	115.7		
			7.5	81.5	4.3	2.3	4.4	83.2	1.9	84.5	-0.5	85.8	-3.6	87.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.1	40.5	43.6	53.0	61.7	69.7	78.8	83.4	82.3	74.4	63.2	52.2	41.1
	DBStd	17.42	10.24	10.92	10.36	8.48	7.74	5.27	5.65	6.15	6.49	8.50	9.29	9.93
	HDD50	1049	323	233	86	13	1	0	0	0	0	7	82	304
	HDD65	3229	760	603	388	157	44	0	0	0	7	138	391	741
	CDD50	5466	29	53	180	364	611	865	1036	1000	733	417	149	29
	CDD65	2171	0	2	17	59	189	415	571	535	290	83	9	1
	CDH74	24863	5	23	152	489	1735	4667	7391	6633	2921	765	79	3
	CDH80	12165	0	3	25	106	596	2206	4138	3635	1233	215	8	0
Wind	WSAvg	9.6	9.7	10.0	11.1	11.7	10.5	9.7	8.6	7.8	7.9	9.0	9.8	9.6
Precipitation	PrecAvg	38.7	1.5	1.9	2.8	3.5	5.3	5.1	3.6	3.3	3.7	3.3	1.9	2.2
	PrecMax	56.3	2.6	4.7	8.1	7.9	11.0	11.7	7.2	8.1	10.0	10.2	6.6	5.9
	PrecMin	26.9	0.1	0.5	0.5	1.0	1.4	1.0	0.7	0.2	0.6	0.8	0.0	0.2
	PrecStd	7.8	0.9	1.2	1.9	1.7	2.5	2.9	2.0	2.2	2.2	2.4	1.9	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.3	78.7	83.8	88.5	93.5	100.0	106.1	106.3	98.8	90.1	81.2	72.6
		MCWB	55.2	59.0	62.3	67.7	71.7	77.2	75.6	74.7	72.9	68.9	64.0	58.0
	2%	DB	67.8	71.8	79.0	82.2	90.2	97.0	101.7	101.9	93.0	85.9	75.4	66.1
		MCWB	53.8	56.4	62.6	66.2	72.5	74.5	74.7	74.4	73.2	68.8	60.8	55.6
	5%	DB	62.8	65.8	73.4	79.4	86.3	93.0	99.3	99.1	90.4	81.6	72.1	62.6
		MCWB	51.1	54.9	60.8	64.5	71.6	74.3	75.2	74.9	72.5	66.5	60.3	52.7
	10%	DB	56.9	61.4	70.3	75.5	82.2	90.4	96.7	95.1	87.8	78.8	67.8	57.0
		MCWB	46.9	52.0	58.8	63.5	70.1	74.6	75.6	75.5	72.5	65.3	58.5	49.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.8	63.5	68.7	73.3	79.2	80.5	80.2	80.1	77.8	75.0	69.3	64.1
		MCDB	66.4	68.6	76.8	82.1	84.5	92.8	93.5	93.9	89.4	82.7	74.5	68.0
	2%	WB	56.3	60.2	66.4	70.7	76.0	78.7	78.5	78.6	76.2	72.7	65.7	59.2
		MCDB	63.7	67.6	74.3	78.5	83.9	91.9	92.8	92.9	87.3	80.2	72.7	63.2
	5%	WB	52.3	56.9	64.0	68.5	74.2	77.4	77.9	77.7	75.1	70.4	62.6	54.2
		MCDB	60.0	64.9	69.7	75.5	82.7	89.4	91.8	91.1	85.4	77.0	67.7	59.2
	10%	WB	48.2	52.5	60.7	66.6	72.3	76.0	77.0	76.5	73.8	67.8	59.7	49.3
		MCDB	58.0	60.6	68.0	73.6	81.1	86.9	90.4	89.4	83.6	75.3	65.7	56.5
Mean Daily Temperature Range	5% DB	MDBR	21.4	21.0	22.0	21.1	19.6	19.4	20.4	20.6	21.3	21.9	21.5	19.4
		MCDBR	29.6	28.4	26.9	24.5	23.0	22.2	23.5	24.4	23.9	24.9	26.6	27.5
		MCWBR	17.4	16.8	15.0	13.0	10.3	8.1	6.4	6.9	8.9	12.0	15.1	17.7
	5% WB	MCDBR	25.8	25.0	20.3	19.8	19.5	19.3	20.0	20.3	19.7	19.0	21.9	22.4
		MCWBR	17.6	17.7	13.2	12.3	10.6	8.8	7.4	7.3	8.4	11.0	15.9	18.2
Clear-Sky Solar Irradiance	taub	0.295	0.300	0.341	0.376	0.413	0.424	0.425	0.433	0.387	0.346	0.314	0.292	
	taud	2.506	2.507	2.416	2.332	2.298	2.312	2.340	2.259	2.366	2.476	2.501	2.518	
	Ebn at Noon	290	301	295	288	277	273	272	268	276	280	280	285	
	Edh at Noon	26	29	35	40	42	41	40	43	36	30	26	24	
All-Sky Solar Radiation	RadAvg	885	1071	1405	1736	1867	2130	2159	1965	1625	1239	933	749	
	RadStd	90	162	89	132	200	205	117	119	147	164	121	85	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page